

Lecture 21: Practice Problems — Comprehensive Solutions Guide

[Jupyter solutions.ipynb](#)
[PDF solutions.pdf](#)
[Questions View Set](#)
[Plot Case 1 Affine Grid](#)
[Plot Case 2 SVM Margin](#)
[Plot Case 3 PCA Spectrum](#)
[Language: English](#) | [हिंदी / Hinglish](#) | [Back to Questions](#) | [Practice Problems Overview](#)

Executive Architecture & Engineering Standards

This document provides production-grade reference solutions, formal mathematical proofs, and executive visual engineering implementations for the 3 industry case studies in **Lecture 21**.

Solution to Case 1: Computer Vision 2D Affine Transformations & Homogeneous Coordinates

1. Mathematical Derivations & Proofs

Task 1: Elementary Homogeneous Matrices

For angle $\theta = 45^\circ = \frac{\pi}{4}$, $\cos(\frac{\pi}{4}) = \sin(\frac{\pi}{4}) = \frac{\sqrt{2}}{2} \approx 0.7071$:

$$R\left(\frac{\pi}{4}\right) = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 1 \end{bmatrix}$$

For horizontal shear with $k_x = 0.5$:

$$Sh(0.5) = \begin{bmatrix} 1 & 0.5 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

For translation vector $\mathbf{t} = [3, -2]^T$:

$$T(3, -2) = \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & -2 & 0 & 0 & 1 \end{bmatrix}$$

Task 2: Matrix Composition & Non-Commutativity

Computing $M_1 = T \cdot Sh \cdot R$: First, multiply $Sh \cdot R$:

$$Sh \cdot R = \begin{bmatrix} 1 & 0.5 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{2}}{4} & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 1 \end{bmatrix}$$

Now left-multiply by T :

$$M_1 = T \cdot (Sh \cdot R) = \begin{bmatrix} \frac{3\sqrt{2}}{4} & -\frac{\sqrt{2}}{4} & 3 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -2 & 0 & 0 & 1 \end{bmatrix} \approx \begin{bmatrix} 1.0607 & -0.3536 & 3.0 & 0.7071 & 0.7071 & -2.0 & 0 & 0 & 1 \end{bmatrix}$$

Now computing $M_2 = R \cdot T \cdot Sh$: Notice that in M_2 , the translation occurs before rotation. The translation vector $[3, -2]^T$ gets rotated by R , causing the translational component to become:

$$R \begin{bmatrix} 3 & -2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2}(3 - (-2)) & \frac{\sqrt{2}}{2}(3 + (-2)) \end{bmatrix} = \begin{bmatrix} \frac{5\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \approx \begin{bmatrix} 3.5355 & 0.7071 \end{bmatrix} \neq \begin{bmatrix} 3 & -2 \end{bmatrix}$$

Hence $M_1 \neq M_2$. In computer vision data augmentation pipelines, changing the order of rotation and translation completely relocates the bounding boxes.

Task 3: Determinant & Area Scaling

For any 2D affine map in homogeneous coordinates:

$$M = \begin{bmatrix} A & \mathbf{t} & \mathbf{0}^T \\ 0 & 0 & 1 \end{bmatrix} \implies \det(M) = \det(A)$$

For M_1 , the linear transformation block is $A = Sh \cdot R$:

$$\det(A) = \det(Sh) \cdot \det(R) = (1 \cdot 1 - 0.5 \cdot 0) \cdot (\cos^2 \theta + \sin^2 \theta) = 1 \cdot 1 = 1$$

Since $|\det(A)| = 1.0$, the combined rotation, horizontal shearing, and translation preserves the total polygonal area exactly (area scaling factor is 1.0).

Task 4: Backward Warping & Inverse Matrix

In forward mapping $\mathbf{x}_{dst} = M \mathbf{x}_{src}$, *rounding floating-point coordinates to integer pixel coordinates creates "holes" (unmapped destination pixels) and collisions. Backward warping* (x_d, y_d) and sampling the source image at:

$$\tilde{\mathbf{x}}_{src} = M^{-1} \mathbf{x}_{dst}$$

Since $M_1 = T \cdot Sh \cdot R$, its analytical inverse is:

$$M_1^{-1} = R^{-1} \cdot Sh^{-1} \cdot T^{-1} = R(-\theta) \cdot Sh(-k_x) \cdot T(-\mathbf{t})$$

Numerically:

$$M_1^{-1} = \begin{bmatrix} 0.7071 & 0.3536 & -1.4142 & -0.7071 & 1.0607 & 4.2426 & 0 & 0 & 1 \end{bmatrix}$$



Solution to Case 2: Support Vector Machine (SVM) Maximum-Margin Hyperplane & Vector Projections

1. Mathematical Derivations & Proofs

Task 1: Normal Vector Orthogonality Proof

Let hyperplane $H = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{w}^T \mathbf{x} + b = 0\}$. Take any two distinct points $\mathbf{x}_a, \mathbf{x}_b \in H$. By definition:

$$\mathbf{w}^T \mathbf{x}_a + b = 0 \quad \text{and} \quad \mathbf{w}^T \mathbf{x}_b + b = 0$$

Subtracting the two equations:

$$\mathbf{w}^T (\mathbf{x}_a - \mathbf{x}_b) = 0$$

Since the vector $(\mathbf{x}_a - \mathbf{x}_b)$ lies entirely within the hyperplane surface H , its dot product with $\mathbf{w}$ is zero. Therefore, $\mathbf{w}$ is orthogonal to every vector in H , proving that $\mathbf{w}$ is the normal vector to the hyperplane.

Task 2: Orthogonal Point-to-Hyperplane Distance Formula

Let $\mathbf{x}_0 \in \mathbb{R}^n$ be any arbitrary point, and let $\mathbf{x}_p \in H$ be its orthogonal projection onto the hyperplane. Because $\mathbf{x}_0 - \mathbf{x}_p$ is parallel to the normal vector $\mathbf{w}$:

$$\mathbf{x}_0 - \mathbf{x}_p = d \cdot \hat{\mathbf{w}} = d \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|_2} \implies \mathbf{x}_p = \mathbf{x}_0 - d \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$$

Since $\mathbf{x}_p \in H$, it satisfies $\mathbf{w}^T \mathbf{x}_p + b = 0$:

$$\mathbf{w}^T \left(\mathbf{x}_0 - d \frac{\mathbf{w}}{\|\mathbf{w}\|_2} \right) + b = 0 \implies \mathbf{w}^T \mathbf{x}_0 + b - d \frac{\mathbf{w}^T \mathbf{w}}{\|\mathbf{w}\|_2} = 0$$

Since $\mathbf{w}^T \mathbf{w} = \|\mathbf{w}\|_2^2$:

$$\mathbf{w}^T \mathbf{x}_0 + b - d \|\mathbf{w}\|_2 = 0 \implies d(\mathbf{x}_0, H) = \frac{\mathbf{w}^T \mathbf{x}_0 + b}{\|\mathbf{w}\|_2}$$

Task 3: Geometric Margin Derivation

For canonical support vectors $\mathbf{x}_+$ and $\mathbf{x}_-$: $\mathbf{w}^T \mathbf{x}_+ + b = +1$ and $\mathbf{w}^T \mathbf{x}_- + b = -1$: *Project the difference vector $(\mathbf{x}_+ - \mathbf{x}_-)$ onto the unit normal vector $\hat{\mathbf{w}}$:*

$$\gamma = (\mathbf{x}_+ - \mathbf{x}_-)^T \frac{\mathbf{w}}{\|\mathbf{w}\|_2} = \frac{(\mathbf{w}^T \mathbf{x}_+ - \mathbf{w}^T \mathbf{x}_-)}{\|\mathbf{w}\|_2} = \frac{(1 - (-1))}{\|\mathbf{w}\|_2} = \frac{2}{\|\mathbf{w}\|_2}$$

Task 4: Analytical Parameter Determination

For support vectors $\mathbf{x}_1 = [1, 2]^T$ (class +1) and $\mathbf{x}_3 = [3, 1]^T$ (class -1): The vector connecting the support vectors is $\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_3 = [1 - 3, 2 - 1]^T = [-2, 1]^T$. The normal vector $\mathbf{w}$ is oriented from -1 to +1, so $\mathbf{w} \propto \mathbf{x}_1 - \mathbf{x}_3 = [-2, 1]^T$. Let $\mathbf{w} = k[-2, 1]^T = [-2k, k]^T$. Setting up the canonical support vector constraint:

$$\mathbf{w}^T \mathbf{x}_1 + b = 1 \implies -2k(1) + k(2) + b = 1 \implies 0 + b = 1 \implies b = 1$$

$$\mathbf{w}^T \mathbf{x}_3 + b = -1 \implies -2k(3) + k(1) + 1 = -1 \implies -5k = -2 \implies k = \frac{2}{5} = 0.4$$

Thus:

$$\mathbf{w}^* = [-0.8 \ 0.4], \quad b^* = 1.0$$

The norm of $\mathbf{w}^*$ is:

$$\|\mathbf{w}^*\|_2 = \sqrt{(-0.8)^2 + (0.4)^2} = \sqrt{0.64 + 0.16} = \sqrt{0.80} \approx 0.8944$$

The optimal margin is:

$$\gamma = \frac{2}{\|\mathbf{w}^*\|_2} = \frac{2}{\sqrt{0.80}} = \frac{2}{0.8944} \approx 2.2361 = \sqrt{5}$$

The decision boundary equation is:

$$-0.8x_1 + 0.4x_2 + 1.0 = 0 \implies x_2 = 2x_1 - 2.5$$



Solution to Case 3: Quantitative Finance Risk & Sensor PCA Eigendecomposition

1. Mathematical Derivations & Proofs

Task 1: Covariance Symmetry & Positive Semi-Definiteness

Given $\mathbf{X}_c \in \mathbb{R}^{N \times p}$, the sample covariance matrix is $\mathbf{\Sigma} = \frac{1}{N-1} \mathbf{X}_c^T \mathbf{X}_c$.

1. Symmetry:

$$\mathbf{\Sigma}^T = \left(\frac{1}{N-1} \mathbf{X}_c^T \mathbf{X}_c \right)^T = \frac{1}{N-1} (\mathbf{X}_c)^T (\mathbf{X}_c^T)^T = \frac{1}{N-1} \mathbf{X}_c^T \mathbf{X}_c = \mathbf{\Sigma}$$

2. **Positive Semi-Definiteness:** For any non-zero vector $\mathbf{v} \in \mathbb{R}^p$:

$$\mathbf{v}^T \Sigma \mathbf{v} = \mathbf{v}^T \left(\frac{1}{N-1} X_c^T X_c \right) \mathbf{v} = \frac{1}{N-1} (X_c \mathbf{v})^T (X_c \mathbf{v}) = \frac{1}{N-1} \|X_c \mathbf{v}\|_2^2 \geq 0$$

3. Since Σ is real symmetric, all eigenvalues are real. Since $\mathbf{v}^T \Sigma \mathbf{v} \geq 0$, for any eigenvector $\mathbf{q}_i$ with eigenvalue λ_i :

$$\mathbf{q}_i^T \Sigma \mathbf{q}_i = \mathbf{q}_i^T (\lambda_i \mathbf{q}_i) = \lambda_i \|\mathbf{q}_i\|_2^2 \geq 0 \implies \lambda_i \geq 0 \quad \forall i$$

Task 2: Lagrange Multiplier Proof for Maximum Variance

We wish to maximize $\mathcal{J}(\mathbf{q}) = \mathbf{q}^T \Sigma \mathbf{q}$ subject to $\|\mathbf{q}\|_2^2 = \mathbf{q}^T \mathbf{q} = 1$. Construct the Lagrangian function:

$$\mathcal{L}(\mathbf{q}, \lambda) = \mathbf{q}^T \Sigma \mathbf{q} - \lambda(\mathbf{q}^T \mathbf{q} - 1)$$

Take the gradient with respect to vector $\mathbf{q}$ and set to zero:

$$\nabla_{\mathbf{q}} \mathcal{L} = 2\Sigma \mathbf{q} - 2\lambda \mathbf{q} = \mathbf{0} \implies \Sigma \mathbf{q} = \lambda \mathbf{q}$$

This is the standard eigenvector equation! Substitute $\Sigma \mathbf{q} = \lambda \mathbf{q}$ into the objective function:

$$\mathcal{J}(\mathbf{q}) = \mathbf{q}^T \Sigma \mathbf{q} = \mathbf{q}^T (\lambda \mathbf{q}) = \lambda \mathbf{q}^T \mathbf{q} = \lambda$$

To maximize $\mathcal{J}(\mathbf{q})$, we must choose the maximum eigenvalue λ_1 , and the optimal projection direction $\mathbf{q}^*$ is the principal eigenvector $\mathbf{q}_1$.

Task 3: Eigendecomposition of the Financial Covariance Matrix

For matrix:

$$\Sigma = \begin{bmatrix} 4.0 & 2.0 & 1.0 & 0.0 & 2.0 & 3.0 & 0.5 & 0.5 & 1.0 & 0.5 & 2.0 & 0.0 & 0.0 & 0.5 & 0.0 & 1.0 \end{bmatrix}$$

Computing eigenvalues via characteristic polynomial $\det(\Sigma - \lambda I) = 0$:

1. $\lambda_1 \approx 5.8679$
2. $\lambda_2 \approx 2.0000$
3. $\lambda_3 \approx 1.2586$
4. $\lambda_4 \approx 0.8735$

Total Portfolio Variance (Trace):

$$\text{Tr}(\Sigma) = 4.0 + 3.0 + 2.0 + 1.0 = 10.0$$

Sum of eigenvalues:

$$\sum_{i=1}^4 \lambda_i = 5.8679 + 2.0000 + 1.2586 + 0.8735 = 10.0000 \quad (\text{Exact trace match!})$$

Proportion of Explained Variance (PEV_k):

- Factor 1 (λ_1): $\frac{5.8679}{10.0} = 58.68$
- Factor 2 (λ_2): $\frac{2.0000}{10.0} = 20.00$
- Factor 3 (λ_3): $\frac{1.2586}{10.0} = 12.59$
- Factor 4 (λ_4): $\frac{0.8735}{10.0} = 8.74$

Cumulative Explained Variance:

- Top 1 Component: 58.68
- Top 2 Components: 58.68
- Top 3 Components: 78.68

Task 4: Dimension Reduction Criterion

To achieve at least 80 explained risk/variance:

- 1 component achieves 58.68
- 2 components achieve 78.68 (close, but strictly under threshold)
- 3 components achieve 91.27 Therefore, $k = 3$ **principal factors** are required to guarantee > 80 risk coverage.

Task 5: Rank-2 Reconstruction & Frobenius Error

Using the top 2 eigenvectors $\mathbf{q}_1, \mathbf{q}_2$:

$$\mathbf{\Sigma}_2 = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^T + \lambda_2 \mathbf{q}_2 \mathbf{q}_2^T$$

The residual matrix is:

$$\mathbf{\Sigma} - \mathbf{\Sigma}_2 = \lambda_3 \mathbf{q}_3 \mathbf{q}_3^T + \lambda_4 \mathbf{q}_4 \mathbf{q}_4^T$$

Since eigenvectors are mutually orthonormal ($\mathbf{q}_i^T \mathbf{q}_j = \delta_{ij}$):

$$|\mathbf{\Sigma} - \mathbf{\Sigma}_2|_F = \sqrt{\lambda_3^2 + \lambda_4^2} = \sqrt{(1.2586)^2 + (0.8735)^2} = \sqrt{1.5841 + 0.7630} = \sqrt{2.3471} \approx 1.5320$$

The low-rank error matches the theoretical spectral lower bound established by the Eckart-Young-Mirsky Theorem.